

Digital Logic Design Lab Report

D Flip-Flop, JK Flip-Flop, 4-bit Register and 4-bit Counter — Verilog Design & Simulation

This report documents the design, implementation, and simulation of four fundamental sequential logic building blocks in Verilog HDL: a D flip-flop, a JK flip-flop, a 4-bit parallel register built from D flip-flops, and a 4-bit synchronous up-counter. Each module was written as a standalone Verilog source file, verified using a dedicated testbench, and simulated to produce waveform outputs confirming correct functional behaviour.

1. Circuit Design

1.1 D Flip-Flop (d_flipflop.v)

The D flip-flop is a single-bit edge-triggered storage element. On every rising edge of the clock, the output Q takes on the value present at the input D, and holds that value until the next clock edge. An asynchronous active-high reset input is included so that the output can be forced to 0 independent of the clock, which is useful for initializing the circuit into a known state.

Behavioural model:

```
always @(posedge clk or posedge rst)
  if (rst) q <= 0;
  else q <= d;
```

1.2 JK Flip-Flop (jk_flipflop.v)

The JK flip-flop extends the D flip-flop concept with two control inputs, J and K, giving four possible modes of operation on each rising clock edge: hold the current state (J=0, K=0), set the output to 1 (J=1, K=0), reset the output to 0 (J=0, K=1), and toggle the output to its complement (J=1, K=1). This toggle behaviour is what makes the JK flip-flop especially useful for building counters.

Truth table implemented in the always block:

J	K	Q(next)	Operation
0	0	Q	Hold
0	1	0	Reset
1	0	1	Set
1	1	$\sim Q$	Toggle

1.3 4-bit Register (register4.v)

The 4-bit register is built by instantiating four D flip-flops in parallel, each sharing the same clock and reset signal. A 4-bit input bus D[3:0] is loaded into the register on the rising clock edge, and the stored value appears on the output bus Q[3:0]. This module demonstrates how single-bit storage elements are composed into wider, word-level storage.

1.4 4-bit Counter (counter4.v)

The 4-bit counter increments its output Q[3:0] by one on every rising clock edge, cycling through the sequence 0000 to 1111 before wrapping back to 0000. It was implemented using JK flip-flops connected in toggle configuration, with each stage's toggle condition derived from the AND of all lower-order bits (ripple/synchronous toggle logic), along with a shared asynchronous reset to clear the count to zero.

2. Simulation Results

Each module was verified using a self-checking or stimulus-driven testbench that applied clock pulses, reset pulses, and input patterns, and monitored the outputs on a simulator waveform viewer (e.g. GTKWave / EDA Playground / ModelSim). The key waveforms observed are summarized below.

2.1 D Flip-Flop

The testbench toggled the D input while the clock ran continuously and asserted reset partway through the run. The waveform showed Q following D exactly one clock edge later, and Q dropping to 0 immediately once reset was asserted, regardless of the clock.

2.2 JK Flip-Flop

The testbench swept J and K through all four input combinations. The waveform confirmed hold behaviour when J=K=0, correct set/reset when only one input was high, and a clean toggle of Q on every clock edge when J=K=1.

2.3 4-bit Register

Different 4-bit values were applied to the D bus on successive clock cycles. The waveform showed Q[3:0] updating to match D[3:0] one clock edge after each change, and all four bits clearing to 0000 when reset was asserted.

2.4 4-bit Counter

With reset released and the clock free-running, the counter output was observed to increment by one on every rising clock edge, producing the sequence 0000, 0001, 0010, ... , 1111, and then wrapping back to 0000. Asserting reset at any point in the count immediately forced the output back to 0000.

3. Observations

- All four modules responded correctly to their respective clock edges, confirming proper synchronous (and, where included, asynchronous reset) behaviour.
- The D flip-flop's output always lagged the D input by exactly one clock period, verifying correct edge-triggered (rather than level-triggered) operation.
- The JK flip-flop's toggle mode ($J=K=1$) was the most useful behaviour for this lab, since it was reused directly to build the 4-bit counter.
- Composing single-bit flip-flops into the 4-bit register showed how word-level storage scales naturally from bit-level building blocks, with all bits updating in lock-step on the same clock edge.
- The counter correctly wrapped around from 1111 back to 0000 without any glitches, and reset took priority over the clock in all modules, confirming the asynchronous reset was implemented correctly.
- No setup/hold violations or unknown ('x') states appeared in the waveforms after the initial reset pulse, indicating the designs are functionally correct for the simulated clock frequency.

4. Conclusion

The lab successfully demonstrated the design and verification of a D flip-flop, a JK flip-flop, a 4-bit register, and a 4-bit counter in Verilog. Simulation waveforms confirmed that each module behaved as expected, and the exercise reinforced how basic sequential elements can be composed to build larger, word-level sequential circuits such as registers and counters.